

Innomatics Research Labs : Data Science with Gen AI Internship

Below "Backend Project 2 (Debugging)" done by Deepak Kaura

Problem Statement -

*Check the attachment. You have a **note taking application** with only one route (i.e. Home Route). This application is broken.*

TASK -

*Kindly **refactor the code** and try to **make sure the application works properly**. Document the bugs while fixing the app.*

MORE DETAILS -

*Home route contains a **'text field'** and a **'button'**. Users can add a note and all the notes will be displayed as an unordered list below the text field on the same page.*

Bugs Identified & Resolution Approach :-

Bug 1: Home Route Did Not Handle GET Requests

- **Issue:**
 - Home route was configured to accept only **POST** requests
 - Browsers load pages using **GET** by default
- **Impact:**
 - Application failed to load on initial access
- **Resolution Approach:**

- Reviewed route configuration
- Enabled both `GET` and `POST` methods
- **Fix Applied:**
 - Updated route to `methods=[ "GET", "POST" ]`

Bug 2: Incorrect Method Used to Fetch Form Data

- **Issue:**
 - Form data was accessed using `request.args`
 - `request.args` works only for GET requests
- **Impact:**
 - Submitted notes were not captured correctly
- **Resolution Approach:**
 - Analyzed HTTP request lifecycle
 - Switched to correct form data handler
- **Fix Applied:**
 - Used `request.form.get("note")`

Bug 3: Form Submission Method Not Defined

- **Issue:**
 - HTML form did not specify a `method`

- Default behavior triggered a GET request
- **Impact:**
 - Backend and frontend request methods were mismatched
- **Resolution Approach:**
 - Ensured form method aligned with backend logic
- **Fix Applied:**
 - Added `method="POST"` in form tag

Bug 4: Empty Notes Added to Notes List

- **Issue:**
 - Empty or `None` values were added when form was submitted without input
- **Impact:**
 - Notes list contained blank entries
- **Resolution Approach:**
 - Implemented basic input validation
- **Fix Applied:**
 - Added condition to append note only if input exists

Bug 5: Notes Not Displayed on Initial Page Load

- **Issue:**

- Template rendering was tied only to POST requests
- **Impact:**
 - Page did not render properly on first load
- **Resolution Approach:**
 - Centralized template rendering for both GET and POST
- **Fix Applied:**
 - Ensured `render_template()` is always returned